

Part A Chapter 10

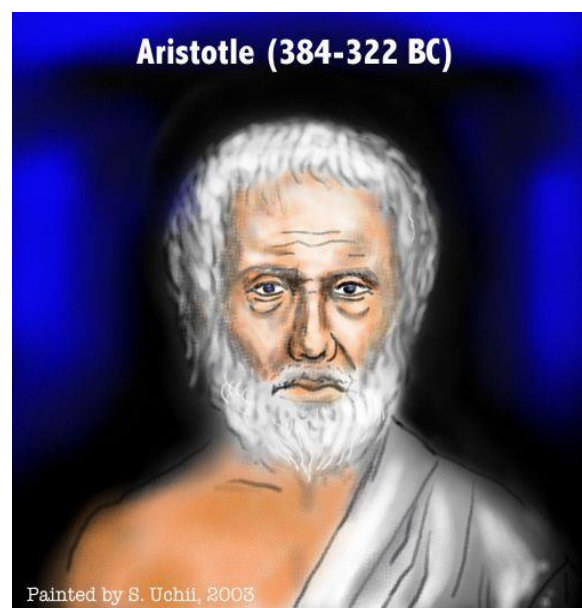
Motivation-Based Behaviour: Beliefs, Desires and Intentions¹

1 Introduction

The third class of behavioural patterns to be discussed is motivation-based behaviour. In this case an animal does not simply respond to stimuli, neither immediately, nor via an internally maintained world state model based on observations in the present and past. In addition to the observations and world state model, *motivational states* play an important role in deciding about its actions. These motivational states can be based on the animal's own character (for example, an animal's character may be that it always wants food), but also on specific aspects of the animal's own state, that have a more dynamic character (such as being hungry). Behaviour can depend on the animal's motivations in several ways. For example, for the set up with the animal, the food and the screen, it may be the case that food is visible, no screen is present and still the animal does not go to the food, because it is not motivated (not hungry). Or, no food has been visible ever, but still the animal goes to various positions to search for food, due to its motivation (goal-driven search behaviour).

1.1 Domain Description and Focus

The model presented in this chapter bases decisions to prepare and perform actions on beliefs and motivations, in particular desires and intentions (e.g., Smith and Mackie, 1999, pp. 316-327, 410-416; Gleitman, 2004, pp. 87-119; Rao and Georgeff, 1991; Jonker, Treur, and Wijngaards, 2003). This model, which sometimes is called the BDI-model (for Beliefs, Desires and Intentions), shows a long tradition in the literature, going back to Aristotle's analysis of how humans (and animals) come to actions; cf. (Aristotle, 350a BC, 350b BC). He shows how the occurrence of certain internal (mental) state properties (e.g., desires) within the living being entail or cause the occurrence



¹ Based on: Jonker, C.M., and Treur, J., Agent-Based Simulation of Animal Behaviour. *Journal of Applied Intelligence*, vol. 15, 2001, pp. 83-115.

of an action in the external world; see also (Nussbaum, 1978). Such internal state properties are sometimes called by him ‘things in the soul’, ‘states of character’, ‘moral states’. In (Aristotle, 350 BC, *Nicomachean Ethics*), this idea is explored in more detail for the case of voluntary human action; for example:

‘We deliberate not about ends but about means. For a doctor does not deliberate whether he shall heal (...). They assume the end and consider how and by what means it is to be attained; and if it seems to be produced by several means they consider by which it is most easily and best produced, while if it is achieved by one only they consider how it will be achieved by this.’ (Aristotle, 350 BC, *Nicomachean Ethics*), Book III, Part 3
(Aristotle, 350 BC, *Nicomachean Ethics*), Book III, Part 5

This describes what today is sometimes called *means-end reasoning*. He explicitly summarises that ‘things in the soul’ (such as sensation, reason and desire) control action:

Now there are three things in the soul which control action and truth - sensation, reason, desire.
(Aristotle, 350 BC, *Nicomachean Ethics*), Book VI, Part 2

Here sensation indicates the sensing of the environment by the organism, which leads, (in modern terms) to internal representations, such as beliefs. Reason indicates that the (rational) choice of a certain action is reasonable to fulfil a given desire. Based on this, Aristotle introduced the following pattern to explain action (practical syllogism):

if	A has a desire D
and	A has the belief that X is an appropriate means to achieve D
then	A will do X

The cognitive model based on beliefs, desires and intentions incorporates such a pattern of reasoning to generate or explain behaviour in a refined form. Instead of a process from desire to action in one step, as an intermediate stage first an intention is generated, and from the intention the action is generated. Thus the process is refined into a two-step process. In this model an action is performed when the subject has the intention to do this action and it has the belief that the opportunity to do the action is there. Beliefs are created on the basis of stimuli that are observed (by sensing). The intention to do a specific type of action is created if there is a certain desire, and there is the belief that in the given world state, performing this action will fulfill this desire (this is the kind of rationality criterion discussed above; e.g., what is called means-end analysis is covered by this). As whether or not a given action is adequate to fulfill a given desire depends on the current world state, this belief may depend on observations or other beliefs about the world state.

Descriptions of behaviour from an external behavioural perspective make use of the input the organism receives (stimuli) and the generated output (behaviour) and their relationships, without involving any internal cognitive state properties. Descriptions of behaviour from a cognitivist perspective, in addition, take internal cognitive state properties such as beliefs,

desires and intentions into account, and their mutual relationships. The status of the internal cognitive state properties involved in such a cognitivist description can be of two types:

(a) *Realist perspective*

The internal cognitive state properties are (known to be) related to certain ‘real’ physical (e.g., neurological) state properties; these physical state properties are also called *realisers* of the internal cognitive state properties.

(b) *Instrumentalist perspective*

The internal cognitive state properties are not (known to be) related to ‘real’ physical state properties: no realisers are assumed. These state properties are an instrument, for example, to make predictions on behaviour, but do not necessarily relate to something existing in reality.

The first, realist perspective makes the dynamic relationships between the internal cognitive state properties correspond to causal relationships, in the physical, chemical, physiological, neurological sense, between their realisers. A description based on the internal cognitive state properties in principle can be replaced by a causal description in terms of the realisers and their causal relationships. For example, an animal damaging its tissue feels pain and subsequently will move away. If having pain can be related to a physical state X, then the explanation ‘moving away because of pain because of tissue damage’ can be replaced by ‘moving away because of physical state X because of tissue damage’, where the ‘because of’ relations in the latter case are based on causal relationships between physical state properties.

As an instance of the *instrumentalist perspective* and opposed to descriptions from a direct physical perspective (the physical stance), in (Dennett, 1987, 1991) the *intentional stance* (or folk-psychological stance) is put forward, which is often used to explain behaviour of others by attributing certain cognitive states to them (within Social Psychology this is called *causal attribution*; see, for example, Smith and Mackie, 1999, pp. 77-98; Gleitman, 2004, 384-387). For example:

Why is he reacting in a negative manner to me?
Because he knows you were with his best friend last night.
How does he know that?
He saw you leaving his place.

Or:

Why is he always bringing chocolate for me?
I believe it is because he wants to date you and he knows you like chocolate.

According to the intentional stance, an individual is assumed to decide to act and communicate based on intentional notions such as beliefs about its environment and its desires and intentions. These decisions, and the intentional notions by which they can be

explained and predicted, generally depend on circumstances in the environment, and, in particular, on the information on these circumstances just acquired by interaction (i.e., by observation and communication), but also on information acquired by interaction in the past. An example put forward by Dennett is:

‘Predicting that someone will duck if you throw a brick at him is easy from the folk-psychological stance; it is and will always be intractable if you have to trace the photons from brick to eyeball, the neurotransmitters from optic nerve to motor nerve, and so forth.’ (Dennett, 1991), p. 42.

As another illustration of this, to function more efficiently in an organisation, it is useful if individuals fulfilling a certain role in the organisation can reason about the possible behaviour of the individuals in other roles, for example using the intentional stance. For example, to an individual functioning within an organisation it may be very helpful to have capabilities to predict in which circumstances certain inappropriate desires or intentions are likely to arise within colleagues, as a basis for their behaviour. This may be quite important, either to avoid the arising of these desires and intentions by preventing the occurrence of circumstances that are likely to lead to them (*manipulation*), or if these circumstances cannot be avoided, by being prepared for the consequences of the intentions (*prediction*). Such capabilities of *anticipatory reasoning* about the behaviour of colleagues in an organisation are quite important for an organisation to function smoothly.

1.2 Characteristic patterns to be explored

In this chapter two types of cases are considered. The first is the experimental setting with the animal, food and screen. The second type of cases concerns functioning in an organisation by recognizing and interpreting other employees’ motivations and behaviours. For these cases the same questions can be addressed as before:

- What is the pattern behind this externally observed behaviour?
- Which concepts are needed to describe this pattern from an external behavioural view?
- Which dynamic relationships characterise the pattern of these traces from the external behavioural view?
- Which assumed internal cognitive state properties can be used to generate this externally observed behaviour?
- What is the pattern of dynamics for these internal cognitive state properties?
- How can these internal cognitive dynamics be characterised by dynamic relationships?

Assume that for the set up as discussed in the previous chapter, the external behaviour traces depicted in Table 1 are observed. Compared to delayed response behaviour, the puzzle to find answers to the above questions is a bit harder this time. For example, in trace 7 after time

points 1, 4 and 5 the animal does not go to p2, while the food is there and the screen is absent; this can be interpreted in the sense that apparently the animal has no motivation at these points in time. This contrasts with traces 1, 5 and 6, where apparently the animal was motivated to go to p2 at time point 5. A question to be answered is under which circumstances the animal has the motivation and under which circumstances it has no motivation.

<i>time trace</i>	<i>time point 0</i>	<i>time point 1</i>	<i>time point 2</i>	<i>time point 3</i>	<i>time point 4</i>	<i>time point 5</i>
<i>trace 1</i>	food screen	no food no screen	food screen	food screen	food no screen	food no screen goto p2
<i>trace 2</i>	no food no screen	food no screen	no food screen	no food no screen	no food no screen	no food no screen
<i>trace 3</i>	no food no screen	no food no screen	food screen	food no screen	food no screen	food no screen
<i>trace 4</i>	food no screen	no food no screen	food screen	food screen	food screen	food screen
<i>trace 5</i>	no food no screen	food no screen	no food screen	food screen	food no screen	food no screen goto p2
<i>trace 6</i>	food no screen	no food no screen	food screen	food screen	food no screen	food no screen goto p2
<i>trace 7</i>	food screen	food no screen	food screen	food screen	food no screen	food no screen

Table 1 Example set of external behavioural traces for motivation-based behaviour

2 Analysis of the Main Aspects and Their Relations

In this section the main concepts and their relations needed for the model are identified.

2.1 Identification of relevant concepts

For the animal behaviour case, from an external perspective as input *observations* are used, in particular of the presence of *food* at a certain position, and of the *screen*. Moreover, at the output, *actions* are *performed*, in particular to *go to* a certain position. Internal concepts used are *beliefs*, storing acquired information, *desires*, and *intentions* for certain actions. Beliefs can be used for information about the presence of food or the screen. In summary, the following concepts are relevant:

External behavioural level

- input: observation (of presence of food or screen)
- output: action (go to p2)

Internal cognitive level

- belief (on presence of food or screen, and reason for an action to satisfy a desire)

- desire (to have food)
- intention (to go to p2)

2.2 Global structure of the model

The model is specified at two levels: the external behaviour level, and the internal cognitive level. A global overall picture of the model can be found in Figure 1.

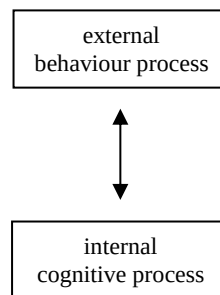


Figure 1 Global structure of the model

Interaction between the levels can take place by transferring input from the behavioural level to the cognitive level, determining output by the cognitive process, and transferring this output back to the behavioural level. Alternatively, the model at the external behavioural level can also be used to determine the outputs from the inputs.

2.3 Relationships between the identified concepts

The relationships between the identified concepts are as follows; see also Figure 1.

Relationships at the external behavioural level for motivation-based behaviour

The traces shown in Table 1 satisfy the following dynamic relationships:

- observation of the absence of the screen affects going to p2
- observation in the past that food at p2 was present, at a later time was not present and after that again was present affects going to p2

Notice that the second relationship is a rather complex temporal relationship involving four different time points; it is not in an executable format, where only a time point for the current state and a time point for the next state are involved. Therefore it cannot be depicted in a graphical manner by arrows and nodes, and cannot directly be used for simulation (from an external behavioural perspective).

Relationships at the internal cognitive level for motivation-based behaviour

Why would the behavioural pattern identified above be in the class of motivation-based behaviour? This question can be answered by showing how the behaviour can be generated

by an internal cognitive model in which motivational concepts play an important role. In this subsection this question is addressed along this line by assuming two motivational internal cognitive state properties: *desire* and *intention*. Moreover, also a belief is relevant: an *additional reason* (to generate an intention from a desire), and an observation of an *opportunity* (to actually perform an intended action).

In Figure 2 the general dynamic relationships between the motivational concepts used are shown. Performing an action depends on the presence of the intention for the action and the belief in an opportunity (which also can be the observation of this fact). The intention depends on the presence of a desire and a belief that the intention is appropriate to get the desire satisfied. A desire can depend on any combination of beliefs and observations over time, or can be directly related to biological states (e.g., a body running out of fuel generates a desire for food).

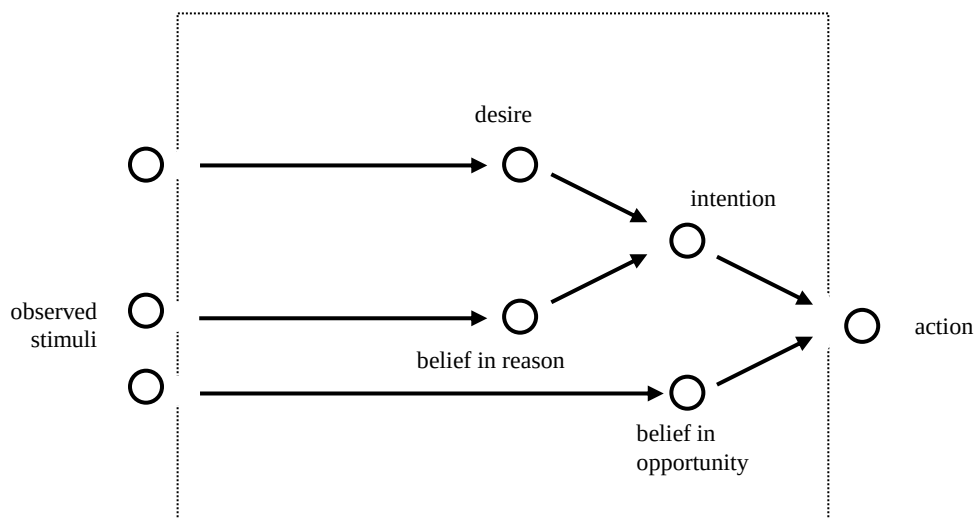


Figure 2 Generic relationships between the motivational concepts

In Figure 3 the relationships for the particular internal cognitive model for the animal behaviour case considered here is shown. The observation of the absence of the screen plays the role of the opportunity here, and the belief that there is food at position p2 plays the role of the reason why the indicated intention to go to p2 is an appropriate choice in the light of the desire for food. For this case the desire is generated as soon as it is noticed that a change occurs in the availability of food because at some time point the observation that no food is present comes in, which differs from the belief that food is present (which is based on an earlier observation): while the animal shows no desire as long as food is available, it starts to desire it when it is not there anymore. Such a pattern of desire generation occurs more often,

in children (e.g., with respect to toys), but also in grown-ups (e.g., with respect to relationships). Sometimes this is summarised as ‘you always want what you cannot get’.

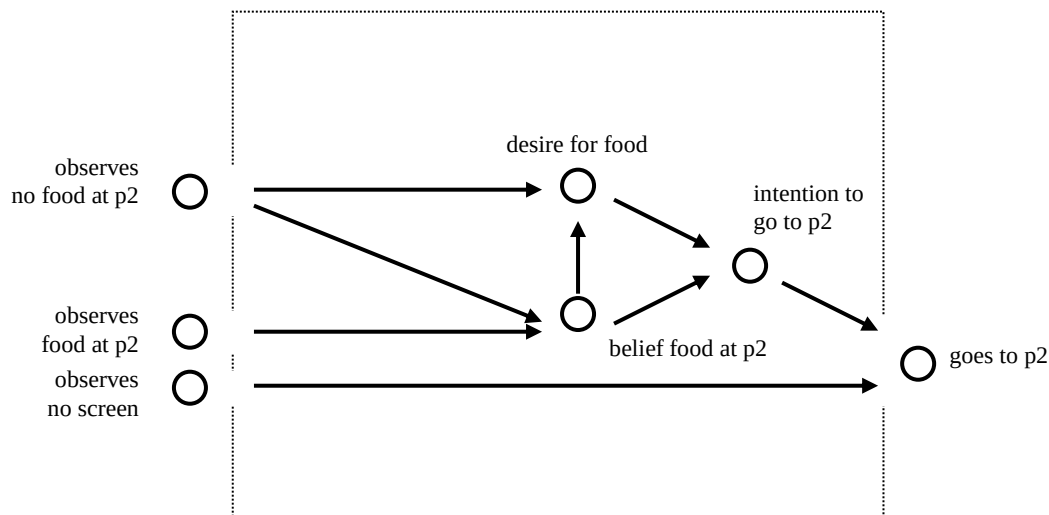


Figure 3 Relationships in the example animal model at the internal cognitive level

3 The Detailed Model

In this section the detailed model is introduced. It is a variation on the models described in (Jonker and Treur, 2007; Jonker, Treur and Wijngaards, 2003). First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To formalise the concepts for each of them a logical atom is introduced; see Table 2. These atoms are based on predicates for observation, performance, belief, desire and intention.

External behavioural level		
observation	(of presence of food or screen)	observed(at_position(food, p2)) observed(screen)
action	(go to p2)	performed(goto(p2))
Internal cognitive level		
belief	(of presence of food or screen)	belief(at_position(food, p2)) belief(screen)
desire	(to have food)	desire(have_food)
intention	(to go to p2)	intention(goto(p2))

Table 2 Formalisation of the relevant concepts

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships, which are analysed in more detail below. They describe the basic parts of the process; see also Figure 3.

3.2.1 Relationships for motivation-based behaviour at the external behavioural level

The following informally expressed dynamic relationship is satisfied by the traces in Table 1:

The animal will go to p2 at each time point that it observes the absence of the screen and in the past the animal observed that food at p2 disappeared, after which again it observed that food was present.

In structured semiformal form:

EMB1

at any time point t1
 if the animal observes the absence of the screen,
 and there exists time points s1, s2, and s3 with $s1 < s2 < s3 \leq t1$ such that
 at s3 the animal observed the presence of food at p2,
 and at s2 the animal observed the absence of food at p2,
 and at s1 the animal observed the presence of food at p2,
 then after time point t1 the animal will go to p2

This type of behaviour is a bit more complex than the previous types of behaviour addressed in Part B, Chapter 2. This more complex behaviour can cope with increased complexity of the environment, for example in environments where food can be absent for longer time periods, so that food-hoarding may be important to survive.

Explanation of motivation-based behaviour from an external behavioural perspective

Given the above description from an external behavioural perspective the following question is addressed.

- What does a behavioural explanation of motivation-based behaviour look like?

The following example explanation illustrates this.

Why does the animal go to p2 ?

The animal goes to p2, because it just observed the absence of the screen, and at an earlier point in time it observed food at p2, whereas at a still earlier point in time it observed the absence of food at p2, and at yet another still earlier point in time it observed the presence of food at p2.

This explanation gets the form of a kind of a historical perspective on the course of events that have led to the current action. It has only reference to causes of the action in an indirect manner (except the observation of the absence of the screen).

3.2.2 Relationships for motivation-based behaviour at the internal cognitive level

Some of the dynamic relations within the generic model for motivation-based behaviour can be specified in structured textual format, according to the following general pattern:

At any point in time
if desire D is present
 and belief B1 is present
then the intention for action A will occur

At any point in time
if the intention for action A is present
 and belief B2 is present
then the action A will be performed

Here it is assumed that belief B1 characterises circumstances under which action A satisfies desire D, and belief B2 characterises circumstances under which there is an opportunity to perform A. Sometimes observations are used for B1 or B2 instead of beliefs. In formal format the dynamic relationships can be expressed as follows:

desire(D) $\wedge$ belief(B1)	$\rightarrow$ intention(A)
intention(A) $\wedge$ belief(B2)	$\rightarrow$ performed(A)

Let's assume for the animal example, that internal cognitive state properties 'belief that food is present at p2', 'desire for food' and 'intention to go to p2' are used with the following dynamic relationships, corresponding to the general relationships formulated above:

IMB1 Desire Generation

at any point in time
if the animal observes the absence of food at p2,
and the belief that food is at p2 holds
then the desire for food will hold

IMB2 Intention Generation

at any point in time
if the desire for food holds,
and the belief that food is present at p2 holds
then the intention to go to p2 will hold

IMB3 Action Generation

at any point in time
if the intention to go to p2 holds,
and the animal observes the absence of the screen
then the animal will go to p2

In formal format, this looks like:

observed(not(at_position(food, p2))) $\wedge$ belief(at_position(food, p2))	$\rightarrow$ desire(have_food)
desire(have_food) $\wedge$ belief(at_position(food, p2))	$\rightarrow$ intention(goto(p2))
intention(goto(p2)) $\wedge$ observed(no_screen)	$\rightarrow$ performed(goto(p2))

In addition, the following dynamic relationships of the belief that food is present at p2 are taken over from Part B, Chapter 2, Section 3.2.5.

IDR1 Belief Generation

for all time points
 if the animal observes that food is present at position p2,
 then the belief that food is present at p2 will hold

IDR4 Belief Persistence

for all time points t1 and t2 with $t1 < t2$
 if the belief that food is present at p2 holds at t1,
 and between t1 and t2 the animal does not observe that food is not present at position p2,
 then the belief that food is present at p2 holds at t2

The internal cognitive state properties desire for food and intention to go to p2 are assumed to persist; for convenience this persistence is assumed unconditional:

IMB4 Desire Persistence

for all time points
 if internal cognitive state property desire for food holds,
 then for every later time point internal cognitive state property desire for food holds

IMB5 Intention Persistence

for all time points
 if internal cognitive state property intention to go to p2 holds,
 then for every later time point internal cognitive state property intention to go to p2 holds

The dynamic relationships **IMB1**, **IMB2**, **IMB3**, **IMB4**, **IMB5**, **IDR1**, **IDR4** defining the internal dynamics are illustrated in a graphical form in Figure 3 above. Notice that the persistence relationships are not visualised. Together these dynamic relationships entail the external dynamic relationship **EMB1**, as is illustrated by the internal simulation traces shown in Section 4.

Explanation of motivation-based behaviour from an internal cognitive perspective

Given the above description from an internal cognitive perspective the following question is addressed.

- What does a cognitive explanation of motivation-based behaviour look like ?

The following example explanation illustrates this.

Why does the animal go to p2 ?

The animal goes to p2, because it just observed the absence of the screen, and it had the intention to go to p2.

Why did the animal have this intention to go to p2 ?

The animal had the intention to go to p2, because it had the desire to go to p2 and the belief that there was food at p2.

Why did the animal have the desire to go to p2; and why did it believe that food was present at p2 ?

The animal desired to go to p2, because it had this desire persisting for a longer time already. The same holds for the belief that food was present at p2.

But then, why did these desire and belief start to be there ?

The desire started to be there because the animal observed that there was no food at p2, and at the same point in time the animal had a belief that food was there.

The belief that food is present at p2 started to be there, because the animal observed the presence of food at p2.

Also this explanation gets the form of a kind of historical perspective on the course of events that have led to the current behaviour. However, in contrast to the explanation in Section 3.2.1, in this case this history can be partitioned according to executable steps. Therefore this explanation has reference to causes in a more direct manner.

4 Simulation Experiments

A number of simulation experiments have been performed. The first one, depicted in Figure 4, illustrates internal cognitive realisation of behaviour a bit similar to what is shown in external behavioural trace 1 from Table 1. Here all internal properties are generated, and also the opportunity is observed. Hence the action goto p2 is generated. Notice that the time scale here is different from the time points in Table 1.

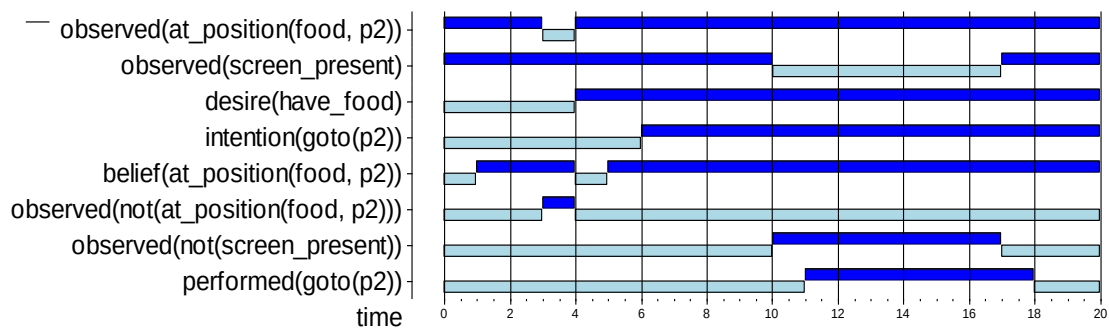


Figure 4 Internal cognitive trace generating external behavioural trace 1 from Table 1

The second internal cognitive trace, which illustrates realisation of behaviour similar to what is shown in external behavioural trace 2 from Table 1 is the one shown in Figure 5. Here the desire is generated, but the intention is not generated, and therefore no goto p2 action occurs.

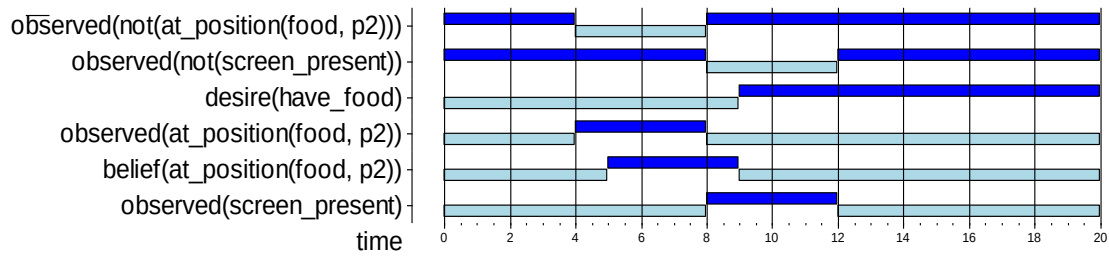


Figure 5 Internal cognitive trace generating external behavioural trace 2 from Table 1

Yet another internal cognitive trace, which illustrates realisation of behaviour similar to what is shown in external behavioural trace 3 of Table 1 is shown in Figure 6. Here the desire is never generated, hence no intention is generated and no action to go to p2 occurs.

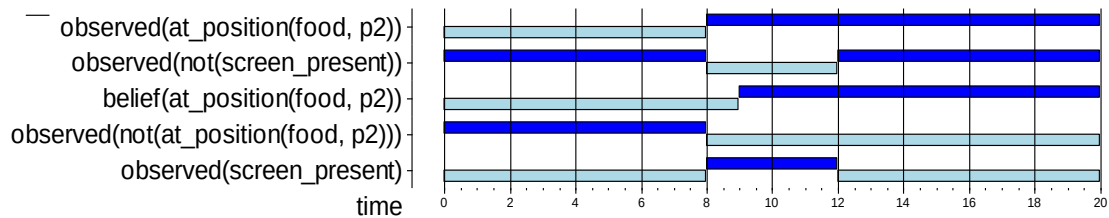


Figure 6 Internal cognitive trace generating external behavioural trace 3 from Table 1

The internal cognitive trace depicted in Figure 7 illustrates realisation of behaviour similar to what is shown in external trace 4. Here eventually all internal cognitive state properties are generated. However, when they are generated the opportunity does not occur anymore, and therefore the intended action is never performed.

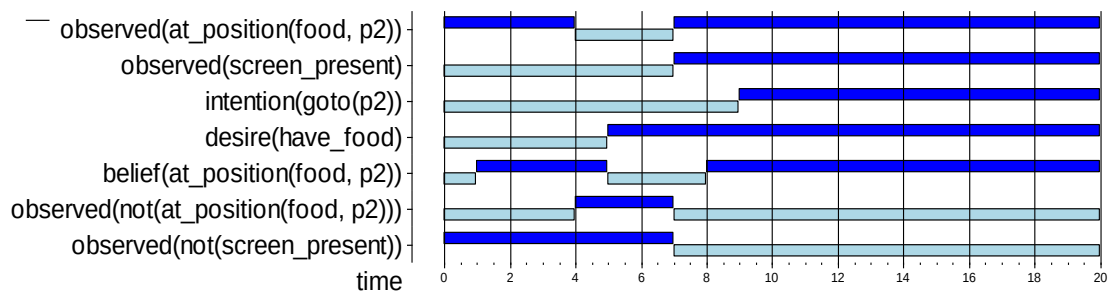


Figure 7 Internal cognitive trace generating external behavioural trace 4 from Table 1

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally some possibilities for variations or refinements of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

A number of different simulations conducted showed the characteristic patterns identified in Section 1.2., and in particular those from Table 1. It is easy to perform a few more experiments to check more scenarios.

5.2 Assumptions made

The following assumptions were made.

- Beliefs persist long time periods.
- Precise timings of the steps were neglected.
- Actions are (only) generated from intentions.
- Intentions are (only) generated from desires
- Actions are (only) performed when an opportunity is there
- An intention for an action is (only) generated when it is believed that the action contributes to satisfaction of the concerning desire.
- Desires and intentions persist forever.
- Desires and intentions have an all or nothing character (no gradual desires or intentions).

Notice that persistence of desires and intentions may easily lead to behaviour that may be considered as unreasonable, for example, in the trace shown in Table 3.

<i>time</i> <i>trace 5</i>	<i>time point 0</i>	<i>time point 1</i>	<i>time point 2</i>	<i>time point 3</i>	<i>time point 4</i>	<i>time point 5</i>	<i>time point 6</i>
<i>input</i>	food screen	no food no screen	food screen	food screen	food screen	no food no screen	no food no screen
<i>internal</i>		belief food	desire food	desire food belief food	desire food belief food intention go to p2	desire food belief food intention go to p2	desire food intention go to p2
<i>output</i>							goto p2

Table 3 Unreasonable internal cognitive trace

5.3 Possible variations and refinements

Variants and refinements can be made for

- Incorporating eat actions and effects of them on desires and intentions. For example, the desire depends on hunger, which occurs after some time period without eating, and disappears after eating.
- Model search behaviour. No beliefs about food are there (for example, because the food is hidden under cups from the start), but due to a desire to have food, intentions are generated to go to positions that were not investigated yet.
- Beliefs decay after some time.
- Incorporating precise timings of the steps.
- Desires and intentions do not persist forever, but are changed when there are good reasons to do so. For example, there is the intention to go for a party, but feeling a bit tired, the desire is not there anymore and intentions are changed
- Desires and intentions have a gradual character.

6 Further Explorations

Some further explorations can be made in the area of mindreading in organisations.

Mindreading: interpretation of behaviour in organisations

Within organisations individuals fulfill certain roles or functions. For these roles, the role behaviour is specified to obtain coherent functioning of the organisation. In human organisations role specifications usually do not completely prescribe behaviours, however. To a certain extent some space of freedom in behaviour and personal initiative is allowed. This freedom has its positive elements; in the first place, human individuals can find more satisfaction and do their work with higher quality if they can do things in their own way. In the second place an organisational structure does not anticipate on all possible circumstances; in unforeseen situations it can be beneficial if individuals have some space to improvise.

The reverse of the medal, however, is that this freedom also may provide possibilities to individuals to show certain behaviours (based on their individual characteristics and interests) that contribute less to the coherence of the organisation. To function more efficient in an organisation, where roles do not completely prescribe behaviour, it is useful if individuals fulfilling a certain role in the organisation can reason in an anticipatory sense about the behaviour of the individuals in other roles, assuming motivational concepts like beliefs, desires, and intentions. In this chapter, some examples of the phenomena described for human organisations are addressed:

(a) An employee has done something wrong, and deliberates whether to tell his manager:

'If he believes that I am the cause of the problems, he will want to fire me.'

(b) An employee has encountered a recurring problem, and knows a solution for this problem, on which he would like to work. He deliberates about how to propose this solution to his manager.

'If I tell this solution immediately he will not believe that the problem is worth working on it. If I make him aware of the problem, and do not tell a solution, he only will start to think himself about it for a while, without finding a solution, and then forget about it. If I make him aware of the problem and give some hints that direct him to a (my) solution, he will believe he contributed to a solution himself and want me to work on it.'

(c) A manager observes that a specific employee in most cases functions quite cooperatively, but shows avoidance behaviour in other cases. In these cases, the employee starts trying to reject the task if he believes that his agenda already was full-booked for the short term, it is not clear to him whether somebody else is not capable of doing the task, and he believes colleagues are available with less full-booked agendas. Further observation by the manager reveals the pattern that the employee shows avoidance behaviour, in particular, in cases that a task is only asked shortly before its deadline, without the possibility to anticipate on getting the task allocated. The manager deliberates about this as follows:

'If I know beforehand the possibility that a last-minute task will occur, I can tell him this in advance, and in addition point out that I need his unique expertise for the task, in order to avoid the behaviour that he tries to avoid the task when it actually comes up.'

The reasoning processes on predicted behaviours described in the examples (as discussed in the introduction) can be based on prescribed role behaviours (as may be the case in (a)), or on an analysis of the other individuals's personal motivations (as is the case in (b) and (c)). To illustrate this, example (c) is addressed by making the following interpretation.

A manager performing mindreading of an employee's mind

The *desire* to avoid a task is created after time t by the employee if the following holds for the history:

- at time t the employee hears the request to perform the task
- at time t the employee observes that the task has to be finished soon
- the employee did not hear of the possibility of the task at any earlier time point

The *intention* to avoid a task is generated after time t if the following holds for the history:

- the desire to avoid the task is available at time t
- the belief that colleagues are capable of doing the task is available at time t
- the belief that colleagues are not full-booked is available at time t

The *action* to avoid the task is generated after time t if the following holds for the history:

- the intention to avoid the task is available at time t
- the belief that the employee's own agenda is full-booked is available at time t

Given this interpretation it can be illustrated how the manager can reason and act in an anticipatory manner to avoid the employee's avoidance desire, intention and/or action to occur. This can be done in the following three manners:

Avoiding the desire to occur

This can be obtained by communicating in advance to the employee that possibly a last minute task will occur. This would make the third condition in the definition of the temporal desire statement fail.

Avoiding the intention to occur (given that the desire occurs)

This can be obtained by refutation of the reason to generate the intention, e.g., by telling the employee that he is the only one with the required expertise.

Avoiding the action to occur (given that the intention occurs)

This can be obtained by refutation of the opportunity, e.g., by taking one of the (perhaps less interesting) tasks from the employee's agenda and re-allocating it to a colleague.

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